TEM Workbench code review

Nov 3, 2026 · @Someone

Scope and data

Reviewed commit: `166cab9` on `main` (3 November 2026). The brief was prepared at `cb4ca63`; the three later commits change overlap/batch code and raise the default FFT minimum separation from 0.12 to 0.50 nm^{-1}. All 152 tests pass at `166cab9` (2 skipped) on Python 3.12.3, NumPy 2.4.4, SciPy 1.17.1, ncempy for DM4 reading. No repository files were changed; experiments ran on copies outside the app.

Evidence comes from two sources:

- **Four real DM4 images** loaded through `Engine.load`: ST1203 (1MX, 0.0120 nm/px, 49 nm field), ST1221 (400kX, 0.0295 nm/px, 121 nm), Ti_0274 (700kX, 0.0168 nm/px, 69 nm), all 4096 × 4096, and ST1144 direct diffraction (0.0115 nm^{-1}/px). Three crop centres per HRTEM image, crops 4–30 nm, titanium HCP + BCC, HCP overall + c/a scale mode.
- **Synthetic lattice images with known answers**, for cases the real set cannot test: known second patterns, known orientations, pure noise.

Limits:

- Real images have no ground truth. Stored labels and the app's own outputs are treated as hypotheses; real-image tests below measure stability and reproducibility, not accuracy.
- ST1156, the crop-size case named in the brief, was not provided. Its results are taken from `docs/implementation-plan.md`.
- The sandbox has 1 CPU, so multi-worker grid throughput was not measured.

Summary

The real images confirm two of the three main synthetic findings and soften the third.

1. **Rank-auto selects the smallest set that one model explains, on real data too.** It kept 4–6 peaks in 35 of 36 real HRTEM runs, against up to 24 in manual mode. It turned 7 manual `insufficient` or `unresolved` results into accepted labels, for example Ti_0274 at 20 nm: BCC from 4 of 24 manual peaks, score 0.69 versus 0.29 in geometry-auto. Its scores are therefore not comparable with manual scores.
2. **Second orientations are never searched (`scale_fitting._seed_candidates`).** Shown on synthetic data, where a dedup-before-truncation patch recovers them. On the four real images the patch changed 1 of 72 labels and found no new secondaries, which

means none of these crops contained a second pattern the old search was hiding, or the detector removed it first. It also lowered 7 scores slightly, because it drops c/a variants of the main orientation that refinement benefits from; the fix needs to keep those.

3. **The detector cutoff follows the strongest peak and rises with crop size.** On real images it rose from 1.80 to 3.33 between 4 and 20 nm. Unlike the synthetic case, real FFTs have many more maxima just below it, and most of them do not recur in an adjacent crop. Excluded maxima recur at 10–20% versus 3–9% for a rotated control, so roughly 1–3 reproducible reflections per crop are lost, among dozens of non-reproducible ones. Lowering the cutoff alone would mostly add noise; selection by reproducibility or local prominence is the better direction.

ST1221 is consistently HCP-compatible and Ti_0274 mostly BCC-compatible across crop sizes and modes; ST1203 is mostly `unresolved`, with BCC at small crops and HCP at one 20 nm crop. These are candidate interpretations only.

Performance: on real images manual takes 0.2–0.6 s, geometry-auto 0.3–0.6 s and rank-auto 0.6–7.1 s per region. A results-preserving cache of weak-peak preparation gave identical output on real crops and saved up to 13%.

What the real images show

Labels across 12 crops per image (3 centres × 4, 8, 12, 20 nm). Zones are the app's 3-index labels.

Image	Manual	Geometry-auto	Rank-auto	Main candidate	Cutoff 4 → 20 nm
ST1221	HCP 9, unresolved 2, insufficient 1	HCP 10, unresolved 2	HCP 10, unresolved 2	HCP [1-10], score 0.61–0.87; best BCC 0.25–0.44	2.03–2.15 → 3.17–3.33
Ti_0274	BCC 7, unresolved 4, insufficient 1	BCC 10, unresolved 2	BCC 12	BCC [1-3-1], score 0.21–0.91	1.80–1.88 → 2.62–2.94
ST1203	BCC 2, HCP 1, unresolved 8, insufficient 1	BCC 3, HCP 1, unresolved 8	BCC 3, HCP 1, unresolved 8	BCC [1-3-1] at 4–8 nm; HCP [1-10] at one 20 nm crop	1.80–1.83 → 2.67–2.92
ST1144 (diffraction, full frame)	HCP [01-1], 0.73	Same	Same	Best BCC 0.21	Not applicable

Observations:

- **Crop-size dependence is real but image-dependent.** ST1221 is stable from 4 to 12 nm and becomes `unresolved` at 20 nm in 2 of 3 centres. Ti_0274's centre 3 is BCC at every size and mode, while centre 1 flips between BCC and unresolved in manual mode. Whether the 20 nm failures come from the detector or from the crop crossing grain or thickness boundaries cannot be separated without manual inspection.
- **ST1203's ambiguity is not a detector artefact alone.** Best HCP and BCC scores are close (often within 0.1) and both low at 12–20 nm. This is the case the implementation plan flags as the stronger BCC candidate; here it is BCC-compatible only in 4–8 nm crops at two centres.
- **Rank-auto shifts scores upward.** In every real crop where both manual and rank-auto returned a primary model, rank-auto's score was equal or higher, by up to 0.20, because it drops the peaks the model does not explain. Comparing rank-auto scores with manual or geometry-auto thresholds overstates support.
- **Diffraction (ST1144):** 9 peaks in the 3–10 nm^{-1} band and the same answer in every mode. The first run took 17.5 s on 1 CPU (detector cost); later modes reused the cached detection and took 0.5–0.6 s.

Prioritized findings

Type: **Defect** = demonstrable implementation error; **Assumption** = scientific choice needing validation; **Perf** = speed only. Evidence marks R for real images, S for synthetic.

#	Type	Finding	Location	Evidence
1	Defect	Rank-auto's objective rewards shrinking the peak set: coverage is divided by the trial's own total weight, so dropping unexplained peaks always helps.	`engine.analyze` rank branch; `core.score_hypothesis`	R: 4–6 peaks in 35/36 runs; 7 manual non-results became accepted labels; scores up to 0.20 higher than manual. S: removed an equal-strength second HCP orientation and every BCC spot in HCP + BCC mixtures.

#	Type	Finding	Location	Evidence
2	Defect	Hypotheses are truncated to the top 3 per c/a ratio before near-duplicates are removed, so a second orientation is never refined.	`scale_fitting._seed_candidates`	S: second HCP [010] orientation with 6 profiles and ~47% of weight never seeded; dedup-first patch recovers it at 0.6× and 0.35×. R: patch changed 1/72 labels (ST1221 20 nm geometry-auto: unresolved → HCP [1-1-2], 0.29) and lowered 7 scores by 0.001–0.022.
3	Defect (likely)	Same truncation in fixed-scale mode: top 2 angles per template at 0.5° rounding.	`analysis_core.search`	Code reading; untested.
4	Assumption	Initial cutoff = max(1.8, 0.35 × strongest enrichment); auto modes can only raise it.	`analysis_core.spectrum`	R: cutoff 1.80 → 3.33 from 4 to 20 nm. Maxima excluded by it recur in an adjacent crop at ~10–20% vs 3–9% chance; those within 0.6 of the cutoff recur at up to 38%. S: genuine second-order spots above every noise maximum were excluded at 20–30 nm.
5	Defect	Weak-maximum detection hard-codes smoothing 0.65 bins and level 1.8, ignoring user detector settings.	`peak_evidence.prepare`	Code reading.

#	Type	Finding	Location	Evidence
6	Assumption	Four matched profiles are required. First-order HCP [0001] and BCC [111] have only 3; at small crops auto modes reach four by admitting noise.	`core.score_hypothesis`, `core.classify`	S: BCC [111] labelled `hcp_compatible` at 4 nm. R: 4 nm crops were `insufficient` in manual at 3 of 9 centres and accepted by auto modes at all three.
7	Assumption	Coverage and validity use the whole peak set as denominator, charging a weaker pattern for not explaining the dominant one.	`core.score_hypothesis`, `core.classify`	Contributes to findings 1 and 2. No real secondaries appeared in 108 runs.
8	Assumption	Zone library: HCP 18 zones, FCC 13, BCC 5 (no [012], [112]). The chance penalty is per template, not corrected for templates tried.	`templates.json`, `crystal_templates`	Counted. Every real BCC label was the same zone, [1-3-1]; an absent zone would surface as a lower-scoring neighbour.
9	Assumption	HCP gets 41 c/a seed ratios and a second scale parameter; cubic gets one.	`scale_fitting.search`, `decomposed_cost`	R: on ST1203 best HCP and BCC scores are often within 0.1, where extra flexibility can decide the label.
10	Defect (minor)	Duplicate removal differs between fixed and auto scale modes.	`core.retain` vs `scale_fitting.retain`	Code reading.
11	Assumption	Ring background is the mean of log power, including the peaks themselves.	`analysis_core.spectrum`	S: median removed 2–3 false maxima at 4 nm; small effect otherwise.

#	Type	Finding	Location	Evidence
12	Assumption	Smoothing in FFT bins suits window-limited peaks.	`analysis_core.spectrum`	S: 1.3 bins pushed genuine weak spots to noise level; broadened-peak test inconclusive. Not swept on real images.
13	Perf	Weak-peak preparation is recomputed for every cutoff level.	`peak_evidence.prepare` via `engine.analyze`	R: 19% of rank-auto time on Ti_0274 20 nm; cached prototype identical, 7.58 → 6.56 s.
14	Perf	Geometry-auto recomputes `core.purity` on the selected set.	`engine.analyze`	Small.

Finding 1 matters most for current outputs: it affects every rank-auto result, including the accepted labels it produces from 4-peak subsets. Findings 2 and 4 matter for weak second patterns, which none of these four real images clearly contains.

Performance breakdown

On real images rank-auto costs 4–8× manual because it repeats full crystal matching at every distinct cutoff, and the number of cutoffs grows with crop size. Manual and geometry-auto stay under 0.65 s per region.

Real HRTEM crops (ST1203, ST1221, Ti_0274; 9 centres), HCP overall + c/a, 1 CPU, single unprofiled run per region. Levels = distinct cutoffs tested by the auto modes.

Crop	Manual median (max)	Geometry-auto median (max)	Rank-auto median (max)	Levels median (max)
4 nm	0.38 s (0.43)	0.38 s (0.55)	1.63 s (4.77)	4 (11)
8 nm	0.35 s (0.39)	0.36 s (0.44)	1.39 s (5.08)	5 (18)
12 nm	0.31 s (0.35)	0.33 s (0.38)	1.88 s (4.87)	6 (20)
20 nm	0.41 s (0.55)	0.42 s (0.62)	3.20 s (7.06)	9 (28)

Direct diffraction (ST1144, full 4096 × 4096 frame): 17.5 s for the first analysis, almost all in detection; 0.5–0.6 s for later analyses reusing it. The implementation plan reports 12.2 → 8.9 s on an M3 Pro with two smoothing threads.

Where rank-auto time goes on a real crop (Ti_0274, 20 nm, 28 levels, cProfile, 12.1 s profiled): hypothesis seeding 41%, peak-evidence inspection with 11 rotated controls 23%, weak-peak preparation 19%, least-squares refinement 14%. FFT and geometry screening are under 1%. Synthetic 65-level cases show the same ordering with seeding at 56%. In Unknown-material mode the orthorhombic fit adds about 60% to every mode (synthetic measurement).

Reuse across nested peak sets

- **Preserves results, measured:** weak-peak candidates depend only on the crop, not on the trial. Cached prototype gave identical JSON on two real Ti_0274 crops and three synthetic ones: 7.58 → 6.56 s (13%) and 2.28 → 2.22 s on real; 9–32% on synthetic.
- **Preserves results, not built:** `prediction_points(template, 1, ratio)` per crop; the pair hypotheses from the 16 strongest peaks, identical for all trials with ≥16 peaks; geometry-auto's selected `purity` result.
- **Changes the search:** inspecting peak evidence only for candidates that can still be retained, warm-starting refinement from the previous level, testing fewer levels. Coarse-to-fine sampling matched rank-auto's full choice in 30/30 synthetic simulations but missed geometry-auto's in 7/30, and rank's agreement comes from its small-set bias.

Grid throughput

Not measurable here (1 CPU). Per-region results are 11–71 KB pickled and numeric threads are pinned to 1 per worker, so CPU-bound regions are the likely limit; the 3.1× four-worker speedup in `docs/parallel-performance.md` is consistent with that. A 100-cell rank-auto grid at 20 nm would take about 5 minutes single-threaded on this sandbox at the measured median.

Validation matrix

Synthetic rows test against known answers; real rows test stability and reproducibility only. Synthetic images: 0.02 nm pixels, Gaussian noise 1.0, amorphous background 0.5.

Scenario	Source	Manual	Geometry-auto	Rank-auto	Note
Clear single crystal, 6+ profiles (HCP [010], BCC [001])	S	Correct 6–30 nm	Correct 4–30 nm	Correct, keeps 4 of 6+ peaks	Agreement 1.0 when labelled
Consistent HCP-compatible image	R: ST1221	HCP at 9/12 crops	10/12	10/12	20 nm unresolved at 2 centres

Scenario	Source	Manual	Geometry-auto	Rank-auto	Note
Consistent BCC-compatible image	R: Ti_0274	BCC 7/12	10/12	12/12	Rank's 12/12 includes 4-peak subsets
Ambiguous image	R: ST1203	Unresolved 8/12	8/12	8/12	Best HCP and BCC often within 0.1
3-profile hexagonal pattern (HCP [0001], BCC [111])	S	`insufficient` at many sizes	BCC [111] → **`hcp_compatible`** at 4 nm	Same	Needs second-order spots
Small crops (4 nm)	R	`insufficient` at 3/9 centres	Accepted at all 3	Accepted at all 3	Same risk as the synthetic row above
Weak second phase (HCP + BCC, 1.0/0.6/0.35×)	S	BCC never accepted	Same	All BCC spots removed	Detector and search both limit
Two orientations of one phase, equal strength	S	`multiple_orientations`	`multiple_orientations`	`hcp_compatible`	Rank bias
Two orientations, 0.6× and 0.35×	S	Second never explored	Same	Same	Fixed by dedup-first patch
Second patterns in real crops	R: 108 runs	None reported	None	None	Patch found none either
Noise only	S: 90 runs	No crystal labels	No crystal labels; up to 70 noise peaks kept	No crystal labels	
Crop-size: cutoff	R and S	1.80 → 3.33 (R), 1.95 → 3.66 (S)	Same pool	Same pool	Auto modes cannot go below it
Peaks just below cutoff	R: 34 crop pairs	Recur in adjacent crop 10–20% (up to 38%) vs 3–9% chance			Some real signal, mostly not
Direct diffraction	R: ST1144	HCP [01-1], 0.73	Same	Same	Auto modes add nothing here

Still needed: ST1156 at 4–30 nm with the old and new detectors side by side; manual inspection of the ST1221 and Ti_0274 20 nm crops that went `unresolved`; and a real region believed to show two orientations of one phase, to test the seed fix on data where it should matter.

Immediate fixes versus experimental alternatives

The real data changed two recommendations: the seed fix must keep c/a variants of the main orientation, and lowering the detector cutoff is no longer an immediate candidate.

Immediate

Change	Results change?	Expected benefit	Risk
Cache weak-peak candidates per crop; filter by trial peaks	No (identical on real and synthetic crops)	13% on real rank-auto, 9–32% synthetic	Cache key must include smoothing, calibration and crop
Cache `prediction_points` per template/ratio; reuse selected `purity`	No	Small, mostly seeding	Low
Seed selection: keep the best c/a variants for each orientation, then add distinct orientations (angle more than ~2° apart) up to the seed limit	Yes, adds candidates	Second orientations recovered (S) without the 0.001–0.022 score losses the first patch caused on real crops	More marginal secondaries; check noise controls
Apply the same fix to the top-2 angles in `core.search`	Yes	Fixed-scale parity	Untested
Label rank-auto results as dominant-pattern fits and stop comparing their scores with manual thresholds, until the objective is changed	No (presentation)	Prevents over-reading 4-peak scores	None
Pass user smoothing and floor into `peak_evidence.prepare`	Only at non-default settings	One consistent signal	Saved non-default results change on rerun
Report when a 3-profile hexagonal pattern blocks fitting	No	Explains `insufficient` results	None
Align duplicate removal between `core.retain` and `scale_fitting.retain`	Yes, minor	Same rule everywhere	Low

Experimental (behind an option, validate first)

Alternative	Suited to	Why	Main risk
Admit sub-cutoff maxima that recur across overlapping or adjacent windows, or that pass a local-prominence test like the existing weak-maximum one	TEM FFT	On real crops excluded maxima carry some reproducible signal but mostly noise; reproducibility separates them where a lower global cutoff would not	Grain boundaries reduce recurrence; needs care in mixed regions
Noise-referenced cutoff from the robust spread of the detector signal	TEM FFT	Fixes the dependence on the strongest peak	On real crops it would admit dozens of non-reproducible maxima; only with the test above
Ring median background	TEM FFT	Removes self-suppression at small crops	Small gain at ≥6 nm
Smoothing combining window width (bins) and intrinsic width (nm^{-1})	TEM FFT	Physically motivated	No supporting measurement yet
Subpixel Gaussian fit with per-peak uncertainty	Both	Matching radius from measured uncertainty	Fit failures on merged peaks
Watershed separation; background-subtracted integrated intensity	Direct diffraction	Spot intensity is meaningful there; FFT power is not	Over-splitting or over-interpreting FFT peaks
Rank-auto objective with a fixed denominator (the full pool, or manual's set) plus a complexity term	Both	Removes the small-set bias found on 35/36 real runs	Rank-auto would select larger sets and accept fewer labels

Alternative	Suited to	Why	Main risk
Pair objective: union of explained weight, shared profiles counted once, chance penalty per pattern, one cost per extra pattern and per extra scale parameter	Both	Principled two-pattern test	Needs calibration on noise and single-crystal controls
Look-elsewhere correction for zones tried per phase	Both	HCP 18 zones vs BCC 5; ST1203′s close HCP/BCC scores are where it matters	Changes phase comparisons

Implementation sequence, benefits and risks

Start with tests, then exact speedups, then one search change at a time. Saved results should stay reproducible under their recorded engine version.

1. **Regression suite.** Synthetic cases from the matrix plus fixed real crops from ST1221, Ti_0274 and ST1203 (store the ROI and expected label as hypotheses). Assert that second patterns are *explored*, not just accepted. Gate: current code fails the synthetic two-orientation cases.
2. **Exact speedups.** Weak-peak cache, prediction-point cache, `purity` reuse. Gate: identical JSON on the suite. Benefit: about 13% on real rank-auto. Risk: stale caches.
3. **Seed fix that keeps c/a variants.** Gate: synthetic second orientations recovered; real labels unchanged except where the old search missed a candidate; no real score drops. Then the same for `core.search`.
4. **Rank-auto presentation.** Mark rank-auto as a dominant-pattern selection in outputs and exports until step 6.
5. **Settings consistency** in `peak_evidence.prepare` and retention rules.
6. **Experimental objectives.** A fixed-denominator rank objective and the pair objective, calibrated on noise-only and single-crystal controls, then compared on the 74-image set from the implementation plan.
7. **Experimental detector.** Reproducibility-gated admission of sub-cutoff maxima. Validate on ST1156 across 4–30 nm and on the 20 nm crops that went `unresolved` here.
8. **Performance follow-ups** once results are stable: evidence pruning, warm starts, coarse-to-fine re-tested under the new objective.

Not recommended: summing top-two ranks, tuning penalties to raise pass rates, lowering the global cutoff on its own, or changing smoothing defaults without the step 7 comparison.

Review scripts (synthetic generator, real-image sweeps, split-half test, cache and seed patches) are in the sandbox, not the repository. I can turn steps 1–3 into a branch on request.